THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
FORM FOUR JOINT EXAMINATION FOR GOVERNMENT SCHOOLS –JEGOS

PRE-MOCK EXAMINATION
033/1 BIOLOGY
MARKING GUIDE 2024

SECTION A (16 Marks)

1.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
D	B	B	D	A	B	C	A	C	E

1pt @ = 10 marks

2.

LIST A	(i)	(ii)	(iii)	(iv)	(v)	(vi)
LIST B	D	A	E	H	I	F

1pt @ = 5 marks

SECTION B (54 Marks)
Answer all questions

3. (a)
   (i) It must be having enough supply of air/must be well ventilated.
   (ii) It must be having electricity.
   (iii) It must be having enough water supplies.
   (iv) It must be having fire extinguishers.
   (v) It must be having preserved specimens. **(Total: 09 marks)**
   (vi) It must be having models.
   (vii) It must be having oven for drying samples.
   (viii) It must be having enough space.
   (ix) It must be having enough heat sources.
   (x) It must be having refrigerator or freezer for storing of specimen.
   (xi) It must be having non-slippery floor.
   (xii) It must be having proper lighting. (1 @ =7 marks)

   (b) A specimen is living organism …for studying biology (example bacteria, grasshopper) while a model is an inanimate object representing a living structure/organism(example plastic human skeleton and bones) ….studying biology (1 @ =2 marks)

4.

(a) **Continuous variation** is the variation, which shows intermediate form between any two extremes, i.e. there is no clear cut distinction between two extremes. Examples of continuous variation are height and weight. Continuous variation is often significantly affected by environmental influences. *02 marks*

Discontinuous variation is the variation which shows a clear cut distinction from one form to another form. Examples of discontinuous variation are ability to roll the tongue, albinism, and blood group. Environment does not influence the characteristics that show discontinuous variation. Example blood group cannot be altered by environment.

02 marks

(b) Causes of variation can be grouped into environment factors and genetic factors:

Environment Factors

- Food: lack of food of a certain diet leads to deficiency diseases such as Kwashiokor and can affect growth. *02½ marks*
- Diseases: diseases in organism can make the individual different from the normal ones.

Genetic factors

- Meiosis: during meiosis there is crossing over and segregation of different gametes. This reduces the chance of individuals being the same.
- Fertilization: during fertilization the nuclei of male and female gametes fuse. This permits parental genes to be brought together in different combinations.
- Mutation- This is a sudden change in gene, which can be inherited. Mutations are caused by mutagens as x rays, cosmic rays, and chemicals such as mustard gas. The individual who has undergone mutation is called a mutant. *Any two points 02½ marks*

5. (a)

Regulation deals with the controlling of the internal environment in organisms, or example, adjusting body temperature …

Homeostasis is an aspect of regulatory processes where organisms maintain a relatively constant or stable internal environment. *01½ marks*

(b)Adaptations:

Insulation by a thick fat layer: Mammals, which live in very cold environments, are <u>insulated by a thick layer of sub-cutaneous fat</u>. Animals such as bears that live in cold regions have thick layer of fat beneath their skin.

Body shapes and sizes: having a smaller surface area to volume ratio increase the ability to conserve heat. Also having short limbs increase the ability to conserve heat.

Other animals without such adaptation can't survive the coldest season unless they migrate or hibernate.

Fur: mammals that live in cold regions have thick fur on the skin which helps in keeping them warm.

Hibernation: Is a state of sleep and greatly reduced metabolic rate that enables certain mammals to survive prolonged periods of low temperature and food scarcity. Examples of animals that hibernate are the polar bears.

Migration: some animals migrate to warmer regions during severe winter times and come back after the winter season is over. *01½ @ = 07½ marks*

(Total: 09 marks)

6. (a)
   - Movement enable animals to find food,without it…..
   - Movemnt enables organims to escape from their enemies, and hence ensuring their survival
   - Movement enables animals to reproduce..

   01 ½ @ = 04½ marks

   (b)
   - Protection of internal organs
   - Support and gives the body shape
   - Facilitates movements
   - Sites for production of blood cells *01 ½ @ = 04½ marks*
   - Storage of minerals

   (Total: 09 marks)

7. (a) Self-pollination is the transfer of pollen grains from the anther to the stigma of the same flower or to the stigma of another flower on the same plant. *01 mark*
   - Advantages of self-pollination: Little pollen is produced because little pollen gets lost.
   - Disadvantages of self-pollination**:** No exchange (mixing) of genetic materials with other plants. No variation hence less chances of survival in a changing environment.
   - Characteristics of self-pollination: it is common in bisexual flowers where stigma and anther ripen at the same time. Usually stigma grows up towards the anther; insects also help. *01@ = 03 marks*

   (b)
   a. Receptacle: This is the swollen tip of pedicel, which forms the flower base to which the floral parts are attached.
   b. Filament: a structure bearing an anther.
   c. Anther: male structure bearing pollen grains. *01@ = 05 marks*
   d. Ovary**:** This is an expanded hollow base, which contains ovules (female eggs).
   e. Style: This is a hollow slander stalk which connects the ovary and the stigma.

(Total: 09 marks)

8. Urine formation involves the following processes:
   - Ultra-filtration (Filtration): pressure builds up in the glomerulus and forces smaller molecules carried in the renal artery into the nephron. Capillary membrane and capsule's membrane act as a filter. The filtrate that enters the nephron is called the glomerulus filtrate. *03 marks*
   - Reabsorption: about 98% of glomerulus filtrate is reabsorbed. All along down the urinary tubule (nephron), cells in the kidney tubule's walls extract useful substances from glomerulus filtrate so that by the time it reaches the collecting ducts a liquid called urine has been formed. Urine contains urea and other waste excretory substances. *02 marks*

- Secretion: secretion involves the introduction of water, mineral, hormones and other substances into the renal tubules by means of the blood capillaries around them. *02 marks*
- Collection/Excretion: This is the process whereby the urine collects in the collecting duct and transported to the urinary bladder for storage before expulsion out of the body. *02 marks*

(Total: 09 marks)

SECTION C (30 Marks)

9. Drug abuse and addiction

Drugs are chemical substances used as medicine, narcotic, or stimulant which bring about changes in the functions of the body. *Drugs have both psychological and physiological effects in human body.* Drugs that affect central nervous system are called **Psychoactive***; they alter consciousness, perception and mood.* *01 mark (introduction)*

Normally drugs are taken to alleviate pains and fight diseases. However, drugs are sometimes abused; **Drug abuse** is the indiscriminate and excessive use of psychoactive drugs for non-medical reasons with no regard to their side effects. *Drug abuse may lead to drug addiction.* **Drug addiction** is the state of dependence caused by regular use of psychoactive drugs (or, compulsive desire to use drugs despite their harmful effect). *02 mark*

Drug addiction (Types of Drug Addiction) comes in Psychological or mental addiction and physical addiction. In psychological or mental addiction, person thinks that his/her well-being depends on the drugs and is unable to stop taking it, i.e. user has an emotional need. In Physical addiction, the body needs the drug in order to work properly. If there is no drug the person suffers withdrawal symptoms. *02 marks*

Withdrawal symptom is the painful reaction when drug is discontinued. *It may cause severe tremors, sweating, and anxiety. It may also progress to physical illness like cramps, muscle pains, nausea vomiting, nausea and convulsion. 01 mark*

Addiction goes together with drug tolerance. **Tolerance** is the condition in which larger doses of drugs are needed to produce the desired results. *01 mark*

People take drugs **(Reasons for taking drugs)** for **medical reasons** such as curing of diseases, desire to **satisfy curiosity** about the effects of drugs, **pressure from peer group** (sense of belongingness to certain group), *desire to have new experience, to escape from reality of life, i.e. hunger, poverty, family quarrels, to draw attention and resist the feeling of neglect, for recreational and excitements purpose…* *Any three 03 marks*

Drug abuse can be controlled by Avoid taking any drug without doctor's prescription, **keeping off from peer pressure** and addicted people, by **being principled** and say no to anyone who offers the drug to you, by **using your leisure time properly** (for sports and hobbies), *by informing community leader or police of any use or trafficking of drugs, by forming counselling club to educate people on how to keep off from drugs (education on proper use of drugs), by seeking help from health officials if you are an addict.*

Any four 04 marks

Conclude *(give a general closing statement about drug addiction)* *01 mark*

Total: 15 marks

10. Introduction: The mammalian heart is a muscular double pump. The right side pumps deoxygenated blood to the lungs, to be oxygenated… 01½ marks

The adaptations of the heart to

- The heart has muscles that are never fatigued.
- The heart has Sino Atrial Node that act as a pacemaker by regulating the heartbeat rates and excitation of the heart.
- There are coronary arteries which lay over the heart surface to provide the heart muscle itself with oxygenated blood. (Any five 02½ @ = 12½
- Has muscular ventricles, with the left ventricle being more muscular to facilitate pumping of the oxygenated blood around the body.
- There are tricuspid and bicuspid valves between the atria and the ventricles, which ensure the blood flows upwards into the major arteries and not back into the atria.
- At the base of the major arteries, there are semilunar valves which prevent the blood returning to the heart as the ventricles relax.
- There is pericardium (pericardial membrane) around the heart that produces lubricating fluid and supports the heart.

The heart is surrounded by a fat layer that acts as a shock absorber

Any relevant conclusion 01 mark **Total: 15 marks**

11. Introduction: Bilharzia is also called schistosomiasis. It is characterized by irritation in the rectum or urinary bladder. These parts become inflamed leading to a discharge of blood in faeces or urine. Damage and inflammation are caused by the escape of eggs from the infected organ to the outside world. *2½ marks*

Cause:
It is caused by schistosomes or blood flukes. *01mark*

Transmission cycle:
By Bilharzia worms. They have two hosts; human and fresh water snail.

- In humans, the adult worms (Mature flukes) live in the walls of the urinary bladder or blood vessels of intestines and rectum.
- They discharge eggs which come out with urine or feces..
- Once in water, the eggs hatch into tiny ciliated larvae.
- The larvae seek appropriate snails and infect them. *3½ marks*
- When a person comes in contact with the water (through bathing or playing in the water), the larvae penetrates the skin and blood vessels.
- In the tissues, the larvae migrate into the liver.
- Mature larvae leave the liver and migrate to the urinary bladder or intestine.

Symptoms
The symptoms of bilharzias include:
-Weight loss.
-Appetite loss. *Any four 01@ = 04 marks*
-Spleen becomes very large.
-Irritation in rectum and urinary bladder.
- Discharge of blood in faeces or urine

Control and prevention.
-Destroy breeding places of snails.
-Keep ducks in water bodies to eat snails.
-Spray molluscicides. *02 marks*
-Educate people on the importance of using toilets.

01marks

Treatment:

Drugs (*the drug niridazole kills one species of Schistosoma but may produce side effects. Praziquantel seems to be effective against all species and has a few side effects)*

Conclusion: *01 mark*

Total: 15 marks